English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 21.01.2024 Test 2

The exact grading formula depends on the results. As an approximation, use 2 plus the points earned. Time allowed: 3 hours. Good luck.

m, n, k denote nonnegative integers.

Problem 1. Let X have distribution function

$$F_X(x) = \begin{cases} 0, & \text{for } x \le 0; \\ ax^2 + bx, & \text{for } 0 < x \le 1; \\ 1, & \text{for } x > 1. \end{cases}$$

1. (0.25 points) Is X a continuous random variable? If so, derive its density.
2. (0.25 points) Find a, b if additionally $\mathbb{P}(X > 0.75) = 0.5$.
3. (0.25 points) If the expectation of a random variable Y exists, then

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y > t)\mathrm{d}t - \int_{-\infty}^0 \mathbb{P}(Y \le t)\mathrm{d}t.$$

   Prove this identity when Y is continuous with density f_Y.

4. (0.25 points) Use the formula to find $\mathbb{E}X$.

Problem 2. Let $U_1, U_2, U_3 \sim U(0,1)$ be independent, and let $U_{(1)}, U_{(2)}, U_{(3)}$ be the values U_1, U_2 and U_3 in increasing order.

For example, if $U_1 = 0.23$, $U_2 = 0.88$, $U_3 = 0.1$, then $U_{(1)} = 0.1$, $U_{(2)} = 0.23$, $U_{(3)} = 0.88$.

1. (0.25 points) Find the densities and expectations of $U_{(1)} := \min\{U_1, U_2, U_3\}$ and $U_{(3)} := \max\{U_1, U_2, U_3\}$.
2. (0.25 points) Is it true that $U_{(2)} \sim U(0,1)$? If yes, prove it; otherwise find its distribution.
3. (0.25 points) Find the variance of $X := -\ln(U_1)$.

Let $X_1, X_2, X_3, X_4 \sim Exp(1)$ be independent.

4. (0.25 points) Find the densities, expectations and variances of $S_2 := X_1 + X_2$, $S_3 := X_1 + X_2 + X_3$ and $S_4 := X_1 + \cdots + X_4$.
5. (0.5 points) Let $Y_k := S_k / S_n$. Prove that $Y_{(k)}$ and $U_{(k)}$ have the same distribution for $k = 1, 2, 3$.

Problem 3.

1. (0.25 points) In the manufacture of specialized automotive parts, the lengths of certain components are measured, and deviations from the recommended length are modelled with mean 0 and variance 9. A component is acceptable if its deviation from the recommended length is less than a threshold s. What percentage of length errors fall within the acceptable range when $s = 1$? Find s such that 95% of components meet the acceptance criterion.
2. (0.5 points) Let $X \sim N(1,1)$. To accuracy 0.01, find $\mathbb{E}|X|$.

Problem 4. The plane is ruled by parallel lines spaced $2t$ apart. A needle of length $2l \le 2t$ is dropped which lands at

a random position.

1. (0.5 points) What is the probability that the needle touches a line?

   Hint: Parametrize by the position of the needle's centre and its angle to the lines.

2. (0.25 points) Suppose $l = t$. What minimum number of independently dropped needles makes the probability that at least half cross a line at least 95%? Explain how and why this procedure can estimate π.
3. *(0.25 points) Bonus: Solve part 1 when $l > t$.*